

Anwar Mousa

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Summary

Data Engineer with hands-on experience building scalable ETL pipelines, real-time streaming systems, and cloud-based data warehouse solutions. Proficient in Apache Airflow, Kafka, Spark, Snowflake, and Python with a strong foundation in data modeling and medallion architecture. Proven track record of optimizing pipeline performance and delivering analytics infrastructure that drives business decisions. Seeking to leverage expertise in distributed data systems and cloud technologies to engineer high-impact data solutions.

Experience

BI Engineer

Jul 2025 – Jan 2026

Link Development · Hybrid

- Designed and delivered end-to-end business intelligence solutions supporting data-driven decision-making across the organization.
- Built and maintained interactive dashboards and executive-level reports for stakeholders using industry-leading BI tools.
- Developed and optimized data models and ETL pipelines to streamline reporting workflows and reduce data latency.
- Collaborated with cross-functional teams to translate complex business requirements into scalable analytical solutions.
- Enforced data quality standards and consistency across all reporting platforms, reducing reporting errors significantly.

Data Engineer

Oct 2024 – May 2025

Digital Egypt Pioneers Initiative (DPEI) · Egypt

- Designed and implemented **ETL pipelines** using **Apache Airflow**, reducing data processing time by **40%**.
- Built real-time streaming data solutions with **Apache Kafka** and **Apache Spark** to handle high-throughput event data.
- Optimized data warehouse queries in **Snowflake**, improving query performance by **35%** through schema redesign and clustering keys.
- Collaborated with cross-functional teams to deliver data-driven insights supporting strategic business decisions.

Data Analyst Intern

Sep 2024 – Oct 2024

NeuronetiX · Hybrid

- Performed statistical analysis on large datasets using **Python** and **SQL** to identify trends and patterns for strategic decision-making.
- Conducted Exploratory Data Analysis (EDA) using **Pandas**, **NumPy**, and **Matplotlib** to extract actionable insights from raw data.
- Created automated reporting dashboards using **Power BI** and **Tableau**, enabling stakeholders to self-serve key metrics.

Education

Bachelor of Computer Science and Informatics

Aug 2022 – Jun 2026

Delta University for Science and Technology · Egypt

- Graduation Project: **Ayn** — AI-native QA & accreditation platform for educational institutions (TypeScript, Next.js, FastAPI, Python) Demo [🔗](#)

Data Analytics Scholarship – Digital Egypt Pioneers Initiative (DEPI)

Oct 2024 – May 2025

Ministry of Communications and Information Technology (MCIT) · Egypt

- Intensive training in data analytics, ETL pipelines, Apache tools, and business intelligence with real-world project delivery.

Projects

Real-Time Forex Data Pipeline & Analytics Dashboard

GitHub [↗](#)

- Built an automated real-time ETL pipeline using **Apache Airflow**, **Apache Kafka**, and **Apache Spark** for processing currency exchange data from Alpha Vantage API.
- Stored and queried processed data in **Snowflake** using a **multi-layer medallion architecture** (Bronze/Silver/Gold) with automated data quality checks.
- Developed an interactive **Streamlit** dashboard featuring candlestick charts, RSI/MACD/EMA indicators, volatility analysis, and custom export options.
- Containerized the full stack with **Docker** for reproducible deployment; implemented secure credential management and structured logging.

Lakehouse Analytics Platform

GitHub [↗](#)

- Architected a modern lakehouse platform implementing the full **medallion architecture** (Bronze/Silver/Gold) for scalable data processing and analytics workloads.
- Leveraged **Apache Spark** and **Delta Lake** for distributed processing and ACID-compliant storage with schema enforcement.
- Containerized using **Docker** for consistent environment management and easy horizontal scaling.

RideTrack – Real-Time Ride-Sharing Analytics

GitHub [↗](#)

- Built a real-time ride-sharing analytics platform processing high-throughput trip data streams using **Apache Kafka** and **Apache Spark**.
- Persisted processed events to **PostgreSQL** and visualized KPIs through a **Streamlit** dashboard for operational monitoring.
- Containerized all services with **Docker** following modern data engineering practices for production-grade deployment.

Real-Time Egypt Economic Indicators (EIS) Pipeline

GitHub [↗](#)

- Designed a comprehensive data pipeline to collect, transform, validate, and analyze Egypt's economic indicators in real time using **Apache Airflow** and **Apache Kafka**.
- Stored clean analytics-ready data in **PostgreSQL**; containerized with **Docker** for reliable orchestration and deployment.

NASA Log Analytics Pipeline

GitHub [↗](#)

- Processed **3.4M+ log entries (61+ GB)** from NASA HTTP server logs (Jul–Sept 1995) using an end-to-end pipeline built on **Apache Airflow**, **Spark**, and **PostgreSQL**.
- Implemented distributed data processing with **Apache Spark** for traffic pattern analysis, error rate monitoring, and resource usage trends.
- Designed interactive monitoring dashboards in **Grafana** to visualize key operational metrics and anomaly detection insights.

Technical Skills

Languages:	Python (Pandas, NumPy, PySpark, Scikit-learn), SQL
Data Engineering:	Apache Airflow, Apache Kafka, Apache Spark, DBT, ETL Pipeline Design
Cloud & Warehousing:	Snowflake, AWS (EC2, S3, RDS), Delta Lake, Data Lakehouse
Databases:	PostgreSQL, MySQL, Microsoft SQL Server
Data Architecture:	Medallion Architecture, Data Modeling, Dimensional Modeling
Visualization:	Streamlit, Grafana, Power BI, Tableau
DevOps & Tools:	Docker, Git, GitHub, Linux, Bash